Traffy Fondue Insight: Labelling Bangkok City Complaints from Text and Tracking How Fast They Get Fixed

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Track Classification (main) + Clustering, with text mining · Rangsit University International College · 28 Oct 2026

1. Problem

Traffy Fondue is a LINE account where Bangkok residents report city problems (roads, flooding, lighting, rubbish). It holds over one million complaints [1]. Users choose the problem category themselves, so many complaints are in the wrong category, and **from November 2025 the category is empty for every complaint**. Fixing speed is also very uneven (under 1 day to 269 days in our sample). Machine learning already sorts public complaints elsewhere [2]; on Traffy, past work sorts complaint text into categories [3][4] or predicts fixing time with one model [5]; nobody combines text labelling, translation and area-level speed analysis.

2. Research question

Can we label Traffy Fondue complaints by problem type from their text, including text translated from other languages, and use those labels to show how fast each type of problem is fixed in each Bangkok district?

- **RQ1 Text labelling:** which model predicts the problem type best, and which complaints were filed in the wrong category?
- **RQ2 Translation:** after Thai–English machine translation, does the model still label complaints correctly?
- **RQ3 Speed and areas:** which types and districts are fixed fast or slow, and which districts group together?

3. Dataset (confirmed)

Source	Traffy Fondue open data, Bangkok Metropolitan Administration. publicapi.traffy.in.th/dump-csv-chadchart/bangkok_traffy.csv
Real / size	Real complaints, over 1 million rows (about 25,000–35,000 per month). Far above the 500-row minimum.
Period	Jun 2022 – Oct 2025 = labelled training data (empty category: 9.8% in 2023, 10.8% in 2024). Nov 2025 – 2026 has no category, so our text model labels it.
Columns	comment (text), type (labels), district, subdistrict, coords, organization, timestamp, last_activity, state, count_reopen, photo. All 16 columns checked.
Cleaning	Remove duplicates and 2021 test rows; split multi-label types; clean Thai text and tokenize; fixing time = last_activity – timestamp for finished cases, extreme values checked.

4. Method (CRISP-DM)

Task	Baseline	Compared models	Evaluation
Text labelling	TF-IDF + Logistic Regression	Linear SVM, Random Forest, WangchanBERTa (vs. published Traffy model [4])	Micro/macro F1, Hamming loss, 5-fold
Translation	Thai text	Same models on Thai→English text (open translation model)	F1 difference Thai vs. English
Fixing speed	Logistic Regression	Random Forest, XGBoost; SHAP to explain	F1, ROC-AUC; train 2022–24, test 2025
Districts	–	k-means, hierarchical clustering	Silhouette score

5. Changes after Milestone 1 and expected contribution

Following Ajarn's feedback, **text labelling is now the main task**, translation lets our team (who cannot read Thai) check the results, and the results feed an **interactive monitoring app** showing problems and fixing speed by district and type. Contribution: labels for the 2025–2026 complaints that have none, a comparison with existing Traffy models, and an area-level view the city can use. Progress, data and files are on our public project page.

6. Next milestones

M3 (10/12 Nov): data cleaned, translated sample, baseline + 2 models with mean ± std. **M4 (24/26 Nov):** speed models, clusters, SHAP, app, IEEE paper (10+ refs), 10-min talk.

[1] BMA PR Office, "Bangkok Celebrates Digital Milestone as Traffy Fondue Resolves One Million City Complaints," Mar. 2026.
[2] F. Caldeira, L. Nunes, R. Ribeiro, "Classification of Public Administration Complaints," SLATE 2022, OASIcs vol. 104, 2022.
[3] A. Pongsathornwiwat, P. Kaewvichit, "Developing Multi-Label Classification Model for Improving Text Categorizing Problems a Case of Traffy* Fondue Platform," Sci. & Tech. Asia, 29(2), 2024.
[4] KDAI-NLP, "wangchanberta-traffy-multi," Hugging Face model. [5] karnjj, NonKhuna et al., "Sed-Young-Wa," GitHub repository, 2023.